

WTG Component-Level CAPEX Modelling — Methodology Report

Tower, Drivetrain/Gearbox, and Blade Landed-Cost Models

1. Purpose and scope

This report documents the calculation methods used in the three interactive component CAPEX tools built for onshore WTG cost-of-supply analysis: **tower**, **drivetrain/gearbox**, and **blade**. Each tool compares landed manufacturing cost across three origins — **China**, **Europe**, **US-domestic** — using a bottom-up cost stack.

Every input is tagged by evidentiary status, consistent throughout:

Tag	Meaning
Sourced	Public, citable data point
Estimate	Editable assumption, no verified public source found
Low-confidence	Single non-primary source, or indirectly implied figure
Data gap	Explicitly not found; left at a neutral default rather than guessed

No component uses an unexplained turbine-cost-vs-MW power law. Where scaling relationships are used, they are tied to a physical load variable (swept area, hub height, rotor radius, torque) with a stated engineering rationale, per NREL's own WindPACT/Cost-and-Scaling methodology.

2. Common cost-stack framework

All three models share the same generic landed-cost structure:

$$C_{\text{landed}} = \left(C_{\text{material}} + C_{\text{labor}} + C_{\text{overhead}} + C_{\text{freight}} \right) \times (1 + m) \times (1 + d)$$

Where:

- C_{material} — material cost, driven by a regionally sourced commodity/material price index
- C_{labor} — fabrication/assembly labor cost
- C_{overhead} — factory overhead

- C_{freight} — logistics cost to the US project site
- m — supplier margin (%)
- d — import duty rate (%), applied only where a component-specific trade measure was verified

The three components differ in **what drives** C_{material} and **how freight is modelled** — this is the analytically important part, not a cosmetic difference.

3. Tower

3.1 Steel mass — engineering scaling, not a power law

Tower steel mass is calculated from NREL's WindPACT baseline scaling relationship, which ties mass to **swept area** and **hub height** — the two variables that set aerodynamic thrust loading and the cantilever bending moment the tower must resist structurally:

$$A_{\text{swept}} = \pi \left(\frac{D_{\text{rotor}}}{2} \right)^2$$

$$M_{\text{tower}} = 0.3973 \times A_{\text{swept}} \times H_{\text{hub}} - 1414 \quad (\text{kg})$$

Where D_{rotor} = rotor diameter (m), H_{hub} = hub height (m).

Source: NREL, *Wind Turbine Design Cost and Scaling Model*, NREL/TP-500-40566 (Fingersh, Hand & Laxson, 2006). — **Sourced**

Validated at the model's own reference point (3 MW / 90 m rotor / 80 m hub $\rightarrow \approx 201$ t), consistent with the order of magnitude of the WindPACT reference tower cost for the same turbine. **Known limitation:** the formula is calibrated to the ~1.5–5 MW class; extrapolating to 6 MW+ modern platforms is outside its validated range.

3.2 Material cost by origin

$$C_{\text{material}} = P_{\text{steel,region}} \times k_{\text{plate}} \times M_{\text{tower}}$$

- $P_{\text{steel,region}}$ — regional HRC price: China \$464/t, Europe \$707/t, US \$942/t (Intratec, Nov 2025). — **Sourced**
- k_{plate} — plate premium multiplier (structural plate vs. commodity HRC), default 1.10 across regions. — **Estimate** (no tower-grade-plate-specific index found)

3.3 Duties

$$d = d_{AD} + d_{CVD}$$

- China AD margin: up to **60.02%**; Vietnam AD margin: up to **58.54%** — 2024 US Commerce/ITC sunset review, utility-scale wind towers. — **Sourced**, but noted as the sunset-review "likely to prevail" margin, not necessarily the exact cash-deposit rate for a named producer.
- CVD applies to China only; Europe/US = 0.

3.4 Full stack

$$C_{\text{total}} = \left(P_{\text{steel}} k_{\text{plate}} M_{\text{tower}} + C_{\text{labor}} + C_{\text{overhead}} + C_{\text{freight}} \right) (1 + m)(1 + d)$$

reported both as \$/tower and normalised to \$/kW.

4. Drivetrain / Gearbox

4.1 Reference cost basis

Unlike the tower, no public mass figures for gearbox/generator/mainframe were recovered — only **cost** references. The baseline is NREL/WindPACT's 3 MW reference-turbine component costs, in 2002 USD:

Component	2002 USD (3 MW)	\$/MW (2002 USD)
Gearbox	\$362,318	\$120,773
Generator (geared/DFIG)	\$195,000	\$65,000
Low-speed shaft + main bearings	\$72,882	\$24,294
Mainframe/bedplate	\$78,129	\$26,043
Mechanical brake	\$5,968	\$1,989

Source: NREL/TP-500-40566, WindPACT reference-turbine cost tables. — **Sourced**

Documented scaling basis used by NREL itself: gearbox **mass** scales with low-speed-shaft **torque** (a physical load variable), and gearbox **cost** is stated as a function of machine rating in kW — which is the basis for the linear-in-MW scaling used here:

$$C_{i,\text{today}} = \left(\frac{C_{i,2002,3\text{MW}}}{3} \right) \times MW \times E$$

Where $E = 2002 \rightarrow \text{today}$ cost escalation multiplier (default $2.6\times$). — **Estimate**, flagged for verification against BLS PPI (machinery) rather than treated as fact.

4.2 Steel-linked components (gearbox, shaft/bearings, mainframe, brake)

$$C_{i,\text{region}} = C_{i,\text{today}} \left[(1 - s_i) + s_i \cdot \frac{P_{\text{steel,region}}}{P_{\text{steel,US}}} \right]$$

Where s_i = steel-linked cost share of component i (editable, default 30–60% depending on component) — **Estimate**. Same regional steel prices as the tower model.

4.3 Generator — copper-linked share

$$C_{\text{generator,region}} = C_{\text{gen,today}} \left[(1 - s_{Cu}) + s_{Cu} \cdot \frac{P_{\text{Cu,region}}}{P_{\text{Cu,US}}} \right]$$

Copper is a **single global LME benchmark** (\$13,335/t, 6 Jul 2026), so $P_{\text{Cu,region}}/P_{\text{Cu,US}} \equiv 1$ for all regions by construction — **Sourced**. This is a deliberate modelling result: unlike steel, the copper-linked share of generator cost carries **no origin cost divergence** from the metal itself; any regional gap must come from labour/fabrication.

4.4 Direct-drive PM variant

$$C_{\text{generator,DD}} = C_{\text{generator,geared}} \times k_{DD}$$

k_{DD} (default $1.8\times$) is an unsourced placeholder — no public NREL coefficient for direct-drive generator cost was recovered. — **Estimate, flagged low-confidence**

Magnet cost (direct-drive only):

$$C_{\text{magnet}} = m_{\text{magnet/MW}} \times MW \times P_{\text{magnet}} \times k_{\text{non-China}}$$

- $m_{\text{magnet/MW}}$ — default 100 kg/MW, derived loosely from a secondary-source range of 200–600 kg/turbine at small-MW scale. — **Low-confidence**
- P_{magnet} — \$60/kg, implied from the US DoD-MP Materials 10-year price floor of \$110/kg being reported as ~\$50/kg above spot (mid-2025). — **Sourced but indirect/implied**
- $k_{\text{non-China}}$ — sourcing-premium multiplier for magnets outside China (default 1.0 China / 1.35 EU,US) reflecting China's ~85–90% share of global NdFeB production (USGS Mineral Commodity Summaries 2025). — **Estimate**, magnitude not independently sourced.

4.5 Duties

No verified component-specific US AD/CVD or Section 301 rate was found for gearboxes or generators, unlike towers. **Data gap** — duty defaults to 0%, explicitly not asserted as "no tariff."

5. Blade

5.1 Mass and cost scaling

Reference anchor: NREL WindPACT 1.5 MW / 70 m-rotor design — single blade mass 4,614.88 kg, single blade cost \$50,477 (2002 USD). — **Sourced** (NREL/TP-500-40566; design description cross-checked in Rinker & Dykes, NREL/TP-5000-67667, 2018).

$$M_{\text{blade}} = M_{\text{ref}} \left(\frac{R}{R_{\text{ref}}} \right)^n, \quad C_{\text{base},2002} = C_{\text{ref}} \left(\frac{R}{R_{\text{ref}}} \right)^n$$

Where R = rotor radius (m), $R_{\text{ref}} = 35$ m, and n = mass-scaling exponent (default 2.5, adjustable range 2.3–2.9). NREL confirms blade mass scales **super-linearly** with radius via curve fits to real design studies — a physically motivated relationship — but the exact published exponent was not recoverable from the sources checked. — **Estimate** (structure is sourced, coefficient is not).

$$C_{\text{base},\text{today}} = C_{\text{base},2002} \times E_{\text{blade}}$$

E_{blade} default 2.3×. — **Estimate**

5.2 Material cost — three-way split

$$C_{\text{material}} = M_{\text{blade}} (f_{gl} P_{gl} + f_r P_r + f_c P_c) + C_{\text{hardware}}$$

- f_{gl}, f_r, f_c — mass shares of glass fibre, resin, and core material (default 45/25/15%, remainder root hardware). — **Estimate**; NREL's own model separately tracks these same four cost buckets (fibreglass, resin & adhesive, core, labour/NAICS), which supports the structure even though the exact weighting is not published.
- P_{gl} — glass fibre price, \$1.50–2.50/kg (India market, used as directional global proxy). — **Low-confidence**
- P_r — epoxy resin price: **Europe \$4.54/kg, Northeast Asia \$1.96/kg** (IMARC, Dec 2025). — **Sourced**, genuine regional divergence, comparable in kind to the tower's steel divergence. No confirmed US-specific figure was found; US value is an interpolated placeholder.

- P_c — core material (balsa/PET), \$4–8/kg. — **Low-confidence**. Over 90% of global balsa originates from Ecuadorian plantations regardless of blade fabrication location — **Sourced** (ZZCHRYSO, 2026) — meaning, structurally, core material behaves like copper in the drivetrain model: a globally-sourced input with limited origin-based price divergence.

5.3 Freight — distance-driven, not flat-rate

$$C_{\text{freight}} = \begin{cases} r_{\text{freight}} \times \text{distance}_{\text{km}} & \text{non-US origin} \\ C_{\text{local}} & \text{US-domestic} \end{cases}$$

Blade logistics are dominated by oversized-cargo costs (escort vehicles, route permitting, road closures) that scale with distance more than with weight — the physical reason blade plants are typically sited near demand regions. r_{freight} and C_{local} are **Estimates**; no public \$/km oversized-cargo index was located.

5.4 Duties

As with the drivetrain, no blade-specific US trade case was found. **Data gap** — defaults to 0%.

6. Sourcing register

Data point	Value	Status	Source
Tower mass scaling formula	$0.3973 \cdot A \cdot H - 1414$	Sourced	NREL/TP-500-40566
Steel price (China/EU/US)	\$464 / \$707 / \$942 per t	Sourced	Intratec, Nov 2025
China/Vietnam tower AD margin	60.02% / 58.54%	Sourced (sunset-review margin)	Federal Register, 89 FR 86314 / 65585 (2024)
Drivetrain component reference costs	See §4.1 table	Sourced	NREL/TP-500-40566
Copper price	\$13,335/t	Sourced	LME via MacroMicro, 6 Jul 2026
China NdFeB magnet share	~85–90%	Sourced	USGS Mineral Commodity Summaries 2025

Data point	Value	Status	Source
Magnet price	~\$60/kg	Sourced but implied	RFF issue brief citing MP Materials-DoD deal, 2025
Blade mass/cost reference	4,614.88 kg / 50, 477(2002)	Sourced	NREL/TP-500-40566; NREL/TP-5000-67667
Epoxy resin price (EU/NE Asia)	\$4.54 / \$1.96 per kg	Sourced	IMARC, Dec 2025
Balsa origin concentration	>90% Ecuador	Sourced	ZZCHRYSO, 2026
Glass fibre / core prices	Ranges, \$5.2	Low-confidence	IndexBox India composites report, 2026
All labor, overhead, freight-rate, plate/margin multipliers	Various	Estimate	Not independently sourced — editable by design
Drivetrain/blade import duties	0% default	Data gap	No verified component-specific trade measure found

7. What this methodology deliberately does not do

- No fitted $\text{Cost} = MW^a$ relationship anywhere in any of the three models.
- No fabricated precision on labour, freight, or duty inputs that lack a public source — these remain user-editable rather than presented as fact.
- No blending of high-confidence sourced figures with low-confidence ones without a visible tag distinguishing them.